Biochemistry Cume
Protein biosynthesis

All questions carry equal points

1. The universal genetic code for an amino acid consists of three bases. Explain why a genetic code in which two bases encode a single amino acid is not adequate for protein synthesis.

2. Puromycin is a well known inhibitor of protein synthesis. How does puromycin function as an inhibitor of protein synthesis.

3. In a 1975 *Nature* paper [*Nature* (1975) **254**, 34-8], two Australian scientists, John Shine and Lyn Dalgarno, identified a nucleotide sequence in the mRNA of prokaryotes. This sequence is now popularly referred to as the Shine-Dalgarno sequence. What is the Shine-Dalgarno sequence? What role does it play in protein synthesis.

4. Unlike bacteria, a virus does not contain elaborate protein synthesis machinery. How does it manage to ensure the synthesis of its proteins?

5. What is the mechanism by which the stop codons signal the termination of synthesis of a protein.